Q1- Write the Lewis dot structure of CO molecule.

Q2- Write the Lewis structure of the nitrite ion, NO_2^- .

Q3- Explain the structure of CO_3^{2-} ion in terms of resonance.

Q4- Explain the structure of CO_2 molecule.

Q5- Explain the formation of a chemical bond.

Q6- Write Lewis dot symbols for atoms of the following elements : Mg, Na, B, O, N, Br.

Q7- Write Lewis symbols for the following atoms and ions: S and S^{2-}; Al and Al^{3+}; H and H^-

Q8- Draw the Lewis structures for the following molecules and ions : H_2S, $SiCl_4$, BeF_2, CO_3^{2-} , HCOOH

Q9- Define octet rule. Write its significance and limitations.

Q10- Write the favourable factors for the formation of ionic bond.

Q11- Discuss the shape of the following molecules using the VSEPR model: $BeCl_2$, BCl_3, $SiCl_4$, AsF_5, H_2S, PH_3

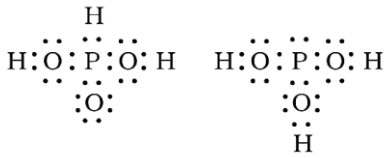
Q12- Although geometries of NH_3 and H_2O molecules are distorted tetrahedral, bond angle in water is less than that of ammonia. Discuss.

Q13- How do you express the bond strength in terms of bond order ?

Q14- Define the bond length.

Q15- Explain the important aspects of resonance with reference to the CO_3^{2-} ion.

Q16- H_3PO_3 can be represented by structures 1 and 2 shown below. Can these two structures be taken as the canonical forms of the resonance hybrid representing H_3PO_3 ? If not, give reasons for the same.



(1) (2)

Q17- Write the resonance structures for SO_3, NO_3^- and N .

Q18- Use Lewis symbols to show electron transfer between the following atoms to form cations and anions : (a) K and S (b) Ca and O (c) Al and N.

Q19- Although both CO_2 and H_2O are triatomic molecules, the shape of H_2O molecule is bent while that of CO_2 is linear. Explain this on the basis of dipole moment.

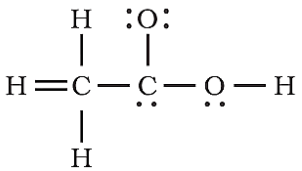
Q20- Write the significance/applications of dipole moment.

Q21- Define electronegativity. How does it differ from electron gain enthalpy ?

Q22- Explain with the help of suitable example polar covalent bond.

Q23- Arrange the bonds in order of increasing ionic character in the molecules: LiF, K_2O, N_2, SO_2 and ClF_3.

Q24- The skeletal structure of CH_3COOH as shown below is correct, but some of the bonds are shown incorrectly. Write the correct Lewis structure for acetic acid.



Q25- Apart from tetrahedral geometry, another possible geometry for CH4 is square planar with the four H atoms at the corners of the square and the C atom at its centre. Explain why CH_4 is not square planar ?

Q26- Explain why BeH_2 molecule has a zero dipole moment although the Be–H bonds are polar.

Q27- Which out of NH_3 and NF_3 has higher dipole moment and why ?

Q28- What is meant by hybridisation of atomic orbitals? Describe the shapes of sp, sp^2, sp^3 hybrid orbitals.

Q29- Describe the change in hybridisation (if any) of the Al atom in the following reaction.
$AlCl_3 + Cl^- \rightarrow AlCl_4^-$

Q30- Is there any change in the hybridisation of B and N atoms as a result of the following reaction?
$BF_3 + NH_3 \rightarrow FB_3NH_3$

Q31- Draw diagrams showing the formation of a double bond and a triple bond between carbon atoms in C_2H_4 and C_2H_2 molecules.

Q32- What is the total number of sigma and pi bonds in the following molecules?
(a) C_2H_2 (b) C_2H_4

Q33- Considering x-axis as the internuclear axis which out of the following will not form a sigma bond and why? (a) 1s and 1s (b) 1s and 2px ; (c) 2py and 2py (d) 1s and 2s.

Q34- Which hybrid orbitals are used by carbon atoms in the following molecules?
(a) CH_3–CH_3; (b) CH_3–CH=CH_2; (c) CH_3-CH_2-OH; (d) CH_3-CHO (e) CH_3COOH

Q35- What do you understand by bond pairs and lone pairs of electrons? Illustrate by giving one exmaple of each type.

Q36- Distinguish between a sigma and a pi bond.

Q37- Explain the formation of H_2 molecule on the basis of valence bond theory.

Q38- Write the important conditions required for the linear combination of atomic orbitals to form molecular orbitals.

Q39- Use molecular orbital theory to explain why the Be_2 molecule does not exist.

Q40- Compare the relative stability of the following species and indicate their magnetic properties;
O_2 O_2^+ O_2^- (superoxide), O_2^{2-} (peroxide)

Q41- Write the significance of a plus and a minus sign shown in representing the orbitals.

Q42- Describe the hybridisation in case of PCl_5. Why are the axial bonds longer as compared to equatorial bonds?

Q43- Define hydrogen bond. Is it weaker or stronger than the van der Waals forces?

Q44- What is meant by the term bond order? Calculate the bond order of : N_2, O_2, O_2^+ and O_2^-